

Assignment 9

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A Bad Data Visualization Example from Real Life

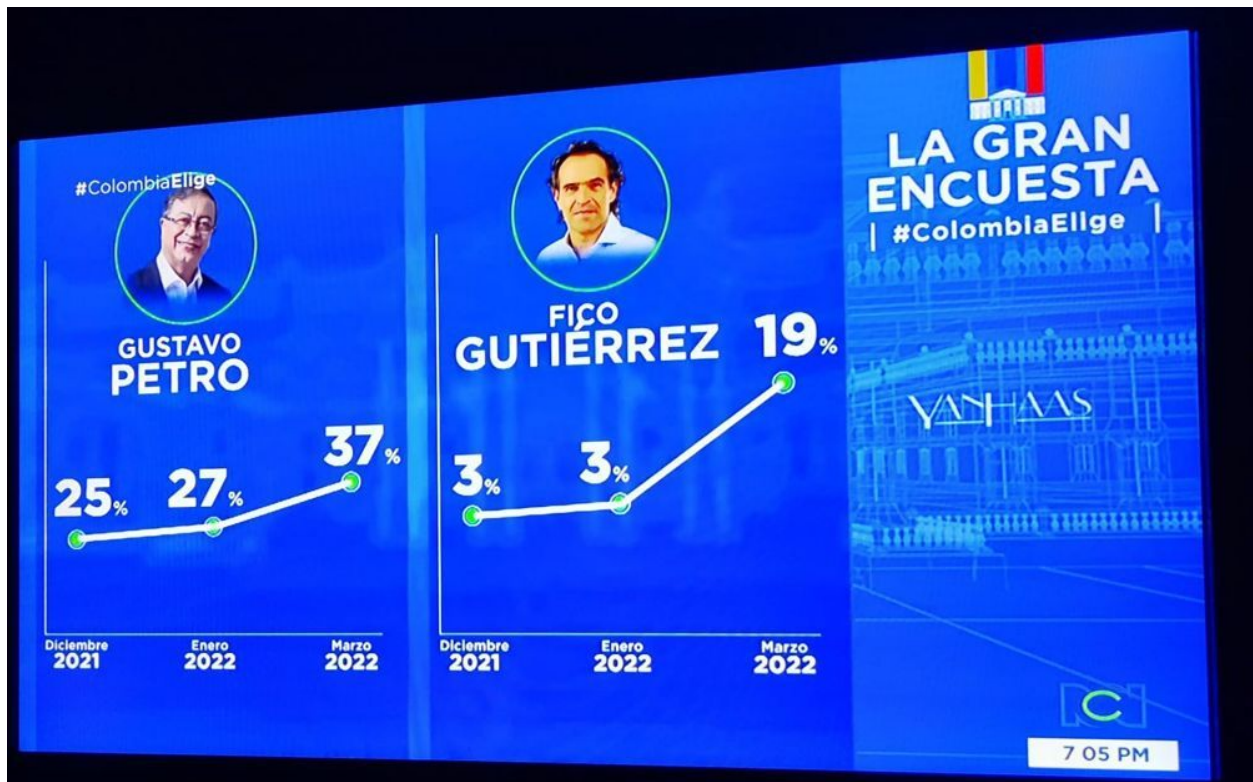


Figure 1. Colombia Elige is a television channel in Colombia that broadcasts coverage of elections and democratic processes.

Explanation of the Problems in the Plot

This plot has several problems that make it hard to understand correctly. First, there is no y-axis, so the viewer cannot clearly see the scale of the values. Even though percentages are written, it is difficult to judge how large the changes really are. Second, the candidates are shown in separate panels, but there is no common scale. This makes comparison misleading, because the increases may look similar even if the actual percentage values are very different. Third, the plot uses line graphs, although the data come from only three survey dates. This gives the impression of a continuous trend, which is not accurate for survey data collected at specific time points. In addition, the steep lines visually exaggerate the changes, especially for Fico Gutiérrez, making the increase appear stronger than it actually is. Overall, these issues reduce clarity and can mislead the viewer. One important issue in this visualization is that it does not clearly state what exactly is being represented. Without a clear explanation, the viewer has to infer that the plot shows the popularity trends of the two political leaders, which creates ambiguity. Another major problem is the use of different vertical

scales across the graphs. Because of this design choice, a lower percentage value in one graph appears visually higher than a much larger percentage value in the other graph. This can easily mislead viewers into thinking that one candidate is more popular than the other, even when the numerical values suggest otherwise. As a result, the visualization does not allow accurate comparison and may distort the viewer's perception of reality. Therefore, despite showing numerical values, the way the data are presented makes this visualization a clear example of bad and misleading data visualization.

Possible Solutions to the Problems in the Plot

The problems in this plot can be solved by making a few simple changes to the design. First, adding a clear y-axis with a common scale would help the viewer understand the size of the changes and allow fair comparison between candidates. Second, instead of showing each candidate in separate panels, the data could be presented in one single chart or in panels with the same scale. This approach would make comparisons more accurate and less misleading. Third, since the data come from survey results at specific time points, using a bar chart or a dot plot would be more appropriate than a line chart. This would avoid giving the impression of a continuous trend. Finally, the graphic could be improved by simplifying the design and adding clearer labels, so the information is easier to read and understand for viewers.

NOTE: Although it wasn't the type of plot we discussed in class as part of bad data visualization, it stood out to me because of its manipulative nature (considering the topics we've been covering for the past two weeks).

References:

[1] <https://www.codeconquest.com/blog/12-bad-data-visualization-examples-explained/>